RECESSION JOBS DASHBOARD

Complete Dashboard Documentation & User Guide

USA · UK · Germany · India | 2008 – 2021 | Version 1.0

201	4	5	4
Records Analysed	Countries	Industries	Recession Periods

1. Purpose & Scope

The Recession Jobs Dashboard is a multi-sheet Excel workbook that provides a structured, evidence-based view of how global economic recessions affect employment, salaries, and workforce composition. It is designed for HR analysts, economists, and business leaders who need to understand labour-market dynamics across geographies, sectors, and company sizes during downturns.

The workbook contains five dashboard views, a raw data sheet, and a pivot-table engine:

- Dashboard — top-level headline KPIs (total layoffs, hiring rate, job-change %, salary impact)
- Industry Dashboard — sector-by-sector comparison of job loss and salary compression
- Trends Dashboard — year-over-year layoff and hiring trends across all four recession windows
- Workforce Dashboard — before/after headcount breakdown by industry
- About — workbook guide and data provenance notes
- Recession Jobs Data — the 201-row source dataset (14 columns)
- pivot_tabels — the pivot-table back-end that powers all dashboard visuals

Design Philosophy

All dashboard figures are driven by pivot tables referencing the 'Recession Jobs Data' sheet. No values are hardcoded on the dashboard tabs themselves — changing or extending the source data automatically refreshes every chart and KPI card.

2. Workbook Structure

Sheet Name	Type	Purpose
Recession Jobs Data	Source Data	Raw 201-row dataset; all metrics originate here
Dashboard	Dashboard	Executive KPI summary — total layoffs, hiring rate, salary change, job loss %
Industry Dashboard	Dashboard	Sector deep-dive: job change %, salary change % and layoff count per industry
Trends Dashboard	Dashboard	Year-over-year layoffs vs. hiring trend lines (2008 → 2021)
Workforce Dashboard	Dashboard	Before/after headcount by industry and aggregate workforce shrinkage
About	Reference	User guide, data dictionary link, and version notes
pivot_tabels	Pivot Engine	Hidden computation layer; pivot tables, GETPIVOTDATA calls, INDEX/MATCH lookups

3. Data Source & Schema

3.1 Source Sheet Overview

Property	Value
Sheet name	Recession Jobs Data
Rows (data)	201 (rows 2–202; row 1 = header)
Columns	14
Years covered	2008, 2009, 2020, 2021
Countries	USA, UK, Germany, India
Industries	Finance, Healthcare, IT, Manufacturing, Retail
Company sizes	Small, Medium, Large
Calculated columns	Job_Change_% and Salary_Change_% are Excel formulas: =(After−Before)/Before

3.2 Column Dictionary

Column	Type	Description	Example
Year	Integer	Recession year of observation	2008, 2009, 2020, 2021
Country	Text	Country of the employer	USA, UK, Germany, India

Column	Type	Description	Example
Industry	Text	Sector classification	IT, Finance, Healthcare …
Company_Size	Text	Organisational size band	Small, Medium, Large
Before_Jobs	Integer	Headcount before the recession period	885, 4223, 1702
After_Jobs	Integer	Headcount after the recession impact	568, 3905, 1393
Job_Change_%	Formula	=(After_Jobs−Before_Jobs)/Before_Jobs (always ≤ 0 in dataset)	−0.358, −0.075
Before_Salary	Currency	Average salary before recession (nominal, local currency)	57,471 / 87,879
After_Salary	Currency	Average salary after recession impact	48,840 / 80,506
Salary_Change_%	Formula	=(After_Salary−Before_Salary)/Before_Salary	−0.150, −0.084
Unemployment_Rate	Decimal	National / sectoral unemployment rate (%)	7.15, 11.41
GDP_Growth	Decimal	GDP growth rate (%) for the country-year	−0.31, 0.53, 2.01
Layoffs	Integer	Employees involuntarily separated during the period	197, 41, 172
Hiring_Rate	Integer	New hires made during the same period	61, 62, 100

4. Dashboard-by-Dashboard Reference

4.1 Dashboard (Executive Summary)

The main Dashboard sheet presents portfolio-level KPIs, giving leaders an at-a-glance view of the overall recession impact across all four periods.

Key metrics displayed:

KPI Card	Dashboard Value	Source (pivot_tabels)
Total Layoffs	28,435	Sum of Layoffs — all years / countries / industries
Total Hires	19,813	Sum of Hiring_Rate — all periods
Avg Job Change %	−13.9 %	Average of Job_Change_% across 201 records
Avg Salary Change %	−9.6 %	Average of Salary_Change_%
Avg Unemployment Rate	7.9 %	Average of Unemployment_Rate

Interpretation

Across all recession periods, the workforce lost approximately 13.9% of jobs on average. Only 69.6 hires occurred for every 100 layoffs (19,813 hires vs. 28,435 layoffs), confirming a net contraction. Salary compression of 9.6% compounded the income impact.

4.2 Industry Dashboard

Compares the five industry sectors across two key outcome measures: average job-change % and average salary-change %. Sectors are ranked from most to least resilient.

Industry	Avg Job Change	Rank (Job)	Avg Salary Change	Rank (Salary)
IT	−11.4 %	**1 (most resilient)**	−8.9 %	**1 (lowest impact)**
Manufacturing	−12.0 %	2	−10.5 %	5
Retail	−14.9 %	3	−9.9 %	4
Finance	−15.6 %	4	−9.4 %	2
Healthcare	−16.2 %	**5 (hardest hit)**	−9.9 %	3
Overall Avg	−13.9 %	—	−9.6 %	—

Key Finding — IT sector resilience

IT consistently showed the smallest job-loss percentage (−11.4%) and the lowest salary compression (−8.9%). This aligns with sustained digital demand even in recessionary periods. Healthcare, despite being essential, experienced the steepest average job cut (−16.2%), driven by budget freezes and funding pressures in public-sector health systems.

4.3 Trends Dashboard

Tracks the year-over-year trajectory of total layoffs and total hires across the four recession windows. The panel enables comparison between the 2008–2009 Financial Crisis and the 2020–2021 COVID shock.

Year	Total Layoffs	Total Hires	Layoffs Δ YoY	Avg Job Change	Avg Hiring Rate
2008	5,429	4,858	Baseline	−14.1 %	110.4
2009	5,941	4,246	+9.4 %	−15.5 %	90.3
2020	8,594	4,985	+44.6 %	−11.5 %	90.6
2021	8,471	5,724	−1.4 %	−14.6 %	106.0

Year	Total Layoffs	Total Hires	Layoffs Δ YoY	Avg Job Change	Avg Hiring Rate
Total	28,435	19,813	—	—	99.1

Key Finding — COVID vs. Financial Crisis

Although the 2020 pandemic produced by far the highest single-year layoff total (8,594), the average job-change % was paradoxically lower than in 2008–2009 (−11.5% vs. −14.1% to −15.5%). This reflects large-company effects: many enterprises held headcount via government furlough schemes while smaller businesses bore concentrated losses. Hiring recovered in 2021 (avg rate 106 vs. 90.3 in 2009), signalling faster post-COVID rebound.

4.4 Workforce Dashboard

Visualises the pre-recession and post-recession headcount by industry, revealing the absolute scale of workforce shrinkage per sector. The dashboard also surfaces the layoff-vs.-hire imbalance by company size.

4.4.1 Before / After Headcount by Industry

Industry	Before Jobs	After Jobs	Jobs Lost	Loss Rate
IT	159,865	145,209	14,656	−9.2 %
Retail	106,011	96,628	9,383	−8.8 %
Manufacturing	97,657	88,910	8,747	−9.0 %
Finance	103,083	91,376	11,707	−11.4 %
Healthcare	90,516	81,204	9,312	−10.3 %
TOTAL	557,132	503,327	53,805	**−9.7 %**

4.4.2 Job Change % by Company Size

Company Size	Avg Job Change %	Interpretation
Medium	−11.6 %	Lowest impact; mid-size firms show most adaptability
Small	−12.7 %	Moderate losses; limited cash buffers extend exposure
Large	−17.0 %	Highest headcount cuts in absolute terms; restructuring-driven

Key Finding — Large Company Disproportionate Cuts

Large organisations recorded the steepest average job-change rate (−17.0%), suggesting that large-scale restructuring and offshoring decisions—more common in big enterprises—amplify headcount reductions beyond what macroeconomic conditions alone would predict. Medium-sized firms (−11.6%) proved most resilient, likely due to greater operational flexibility.

5. Geographic Breakdown

The dashboard enables country-level slicing. The table below summarises total layoffs and relative share for each country derived from pivot_tabels.

Country	Total Layoffs	Share of Total	Total Hires	Net Workforce Δ
USA	9,475	33.3 %	—	Largest layoff volume
UK	8,703	30.6 %	—	2nd highest
Germany	6,978	24.5 %	—	Steepest job-change %
India	3,279	11.5 %	—	Lowest layoff count
TOTAL	28,435	100 %	19,813	Net: −8,622

Note on Country Comparison

Layoff counts reflect the number of records in the dataset attributed to each country and do not represent absolute national layoff tallies. The dataset is structured to enable relative comparisons, not population-level labour statistics. Salary fields are in nominal local currency and are not adjusted for purchasing power parity (PPP).

6. Pivot Table Engine (pivot_tabels sheet)

The pivot_tabels sheet is the computational backbone of the workbook. All dashboard KPI cards and chart series pull values from this sheet using GETPIVOTDATA() and INDEX/MATCH formulas. It should not be edited manually.

6.1 Pivot Tables Defined

Anchor Cell	Measure	Purpose
A1	Sum of Layoffs	Grand total layoffs across all dimensions — feeds main KPI card
A5	Avg of Salary_Change_%	Portfolio salary-compression rate — feeds Salary KPI
A20	Avg of Unemployment_Rate	Portfolio unemployment rate — feeds Unemployment KPI
D2	Avg Job Change by Industry	Feeds Industry Dashboard bar chart (ranked list)
G3	Layoffs + Hires by Year	Feeds Trends Dashboard dual-axis chart
N2	Layoffs by Country	Feeds geographic breakdown panel
S2	Job Change by Company Size	Feeds Workforce Dashboard size-band chart
A50	Avg Job Change by Year	Feeds Trends year-over-year line
A60	Before/After Jobs by Industry	Feeds Workforce before-after clustered bar chart
A87	Avg Hiring Rate by Year	Feeds Trends hiring-rate line series

6.2 Formula Patterns Used

- GETPIVOTDATA("Layoffs", A1) — pulls a specific aggregation from a pivot by field name and anchor cell
- =GETPIVOTDATA("Job_Change_%", A50, "Year", 2008) — parameterised pull with a filter dimension
- =INDEX(A10:A14, MATCH(MAX(B10:B14), B10:B14, 0)) — dynamic label lookup; identifies the industry with the least job loss
- All Job_Change_% and Salary_Change_% columns in the source data are Excel formulas (=(F2-E2)/E2 pattern) — not hardcoded

7. Data Quality & Known Issues

7.1 Missing Values

Four rows (rows 202–205 in the source sheet) contain null Year values and blank numeric fields. These records are excluded from pivot aggregations automatically but remain in the raw data sheet for reference. The effective analytical population is 201 records.

7.2 Currency Heterogeneity

Salary figures (Before_Salary, After_Salary) are denominated in local currency for each country. No PPP or FX normalisation has been applied. Cross-country salary comparisons therefore reflect nominal differences only and should not be used for international pay benchmarking without further adjustment.

7.3 Outlier Records

Field	Anomalous Value	Guidance
Unemployment_Rate	Max: 1,585 %	Exceeds any real-world national rate; likely a unit-scale or data-entry error. Exclude from unemployment trend analysis.
GDP_Growth	Min: −208.7 %	Beyond plausible GDP contraction bounds. Verify source; treat as suspect until validated.
After_Jobs	Max: 503,327	Aggregate figure from Workforce Dashboard pivot, not a single-firm value. No action needed.
Layoffs	Max: 28,435	Grand total across all records; not a single-employer figure. No action needed.

7.4 Recommendations for Extended Analysis

- Use median Job_Change_% (≈ −9.1 %) as the central tendency metric rather than the mean (−13.9 %), which is skewed by extreme records.
- Apply IQR-based outlier filtering to Unemployment_Rate and GDP_Growth before regression or correlation analysis.
- Normalise salary data to USD (or PPP-adjusted) before any cross-country compensation comparison.
- Validate the four null-Year records with the original data source before deciding whether to impute or discard them.

8. How to Use the Dashboard

8.1 Filtering & Slicing

Each dashboard sheet contains Excel Slicers (visible in the workbook UI) linked to the pivot tables on pivot_tabels. Use the slicers to filter by:

- Year — isolate a specific recession window (2008, 2009, 2020, 2021)
- Country — compare a single geography against the portfolio
- Industry — focus on one sector's performance
- Company_Size — analyse Small / Medium / Large firm cohorts independently

Tip

Hold Ctrl (Windows) or Cmd (Mac) while clicking slicer buttons to apply multi-select filters, e.g. selecting both 2008 and 2009 to view the complete Financial Crisis window as a combined period.

8.2 Refreshing Data

- If source data is added or edited in 'Recession Jobs Data', right-click any pivot table on pivot_tabels and select Refresh All to propagate changes to all dashboards.
- Alternatively, go to Data → Refresh All from the Excel ribbon.
- Charts linked to pivot tables will update automatically after refresh.

8.3 Adding New Records

- Append new rows directly below row 202 in the 'Recession Jobs Data' sheet.
- Ensure the Year column contains one of the four recession years (2008 / 2009 / 2020 / 2021) or expand pivot Year filters if adding a new period.
- Job_Change_% and Salary_Change_% columns use relative formulas — copy cell G202 and J202 down into the new rows to auto-calculate.
- Refresh all pivot tables after appending new data.

9. Glossary

Term	Definition
Before_Jobs	Total headcount employed within the firm/sector prior to the recession observation window
After_Jobs	Remaining headcount measured at the end of the recession impact period
Job_Change_%	(After_Jobs − Before_Jobs) / Before_Jobs; negative values indicate net job loss
Before_Salary	Average compensation before recession in nominal local currency
After_Salary	Average compensation after recession in nominal local currency
Salary_Change_%	(After_Salary − Before_Salary) / Before_Salary; salary compression metric
Layoffs	Count of involuntary separations (redundancies/retrenchments) during the period
Hiring_Rate	Count of new employees onboarded during the same recession window
Unemployment_Rate	Percentage of the active labour force without employment at time of observation
GDP_Growth	Year-over-year GDP growth rate (%) for the country in the given year

Term	Definition
Recession Period	One of four economic contractions studied: Global Financial Crisis (2008, 2009) and COVID-19 (2020, 2021)
GETPIVOTDATA	Excel function used to extract a specific value from a pivot table by field name and optional filter dimensions
Slicer	Interactive filter control in Excel that lets users select dimension values to filter pivot tables and charts

This document was generated automatically from the Recession Jobs Dashboard workbook. For questions, data corrections, or dashboard enhancements, contact your Analytics team.